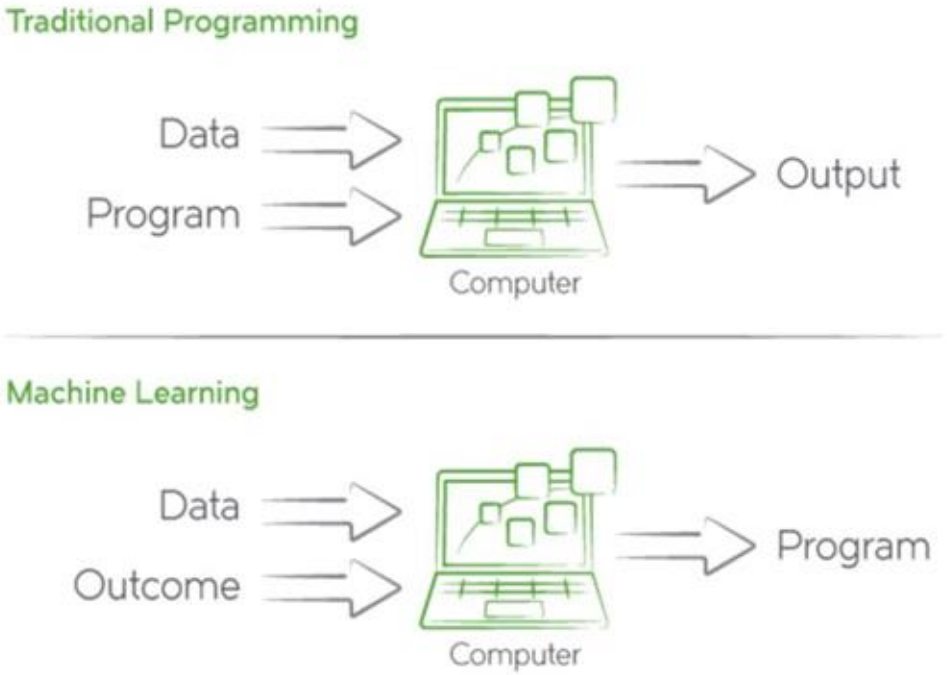
INT3405 - Machine Learning

Lecture 2: General Concepts for ML

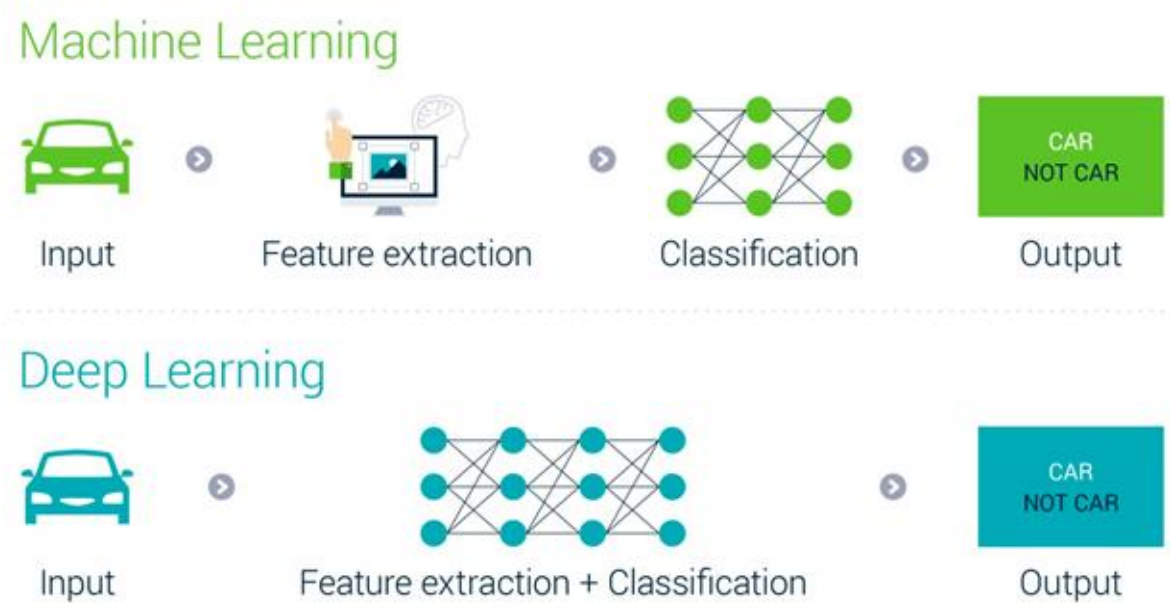
Hanoi, 02/2025

Recap: Traditional Programming vs Machine Learning



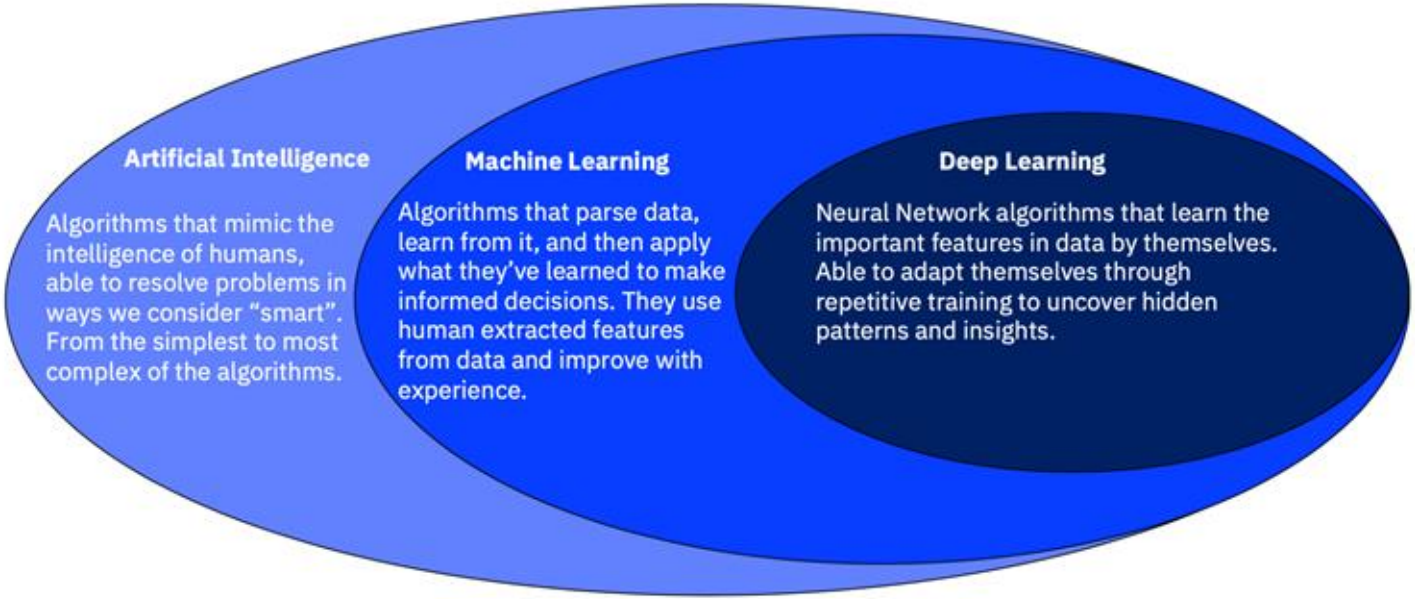
https://images.techhive.com/images/article/2017/05/traditional-programming-vs-machine-learning-100723299-large.jpg

Recap: Machine Learning vs. Deep Learning



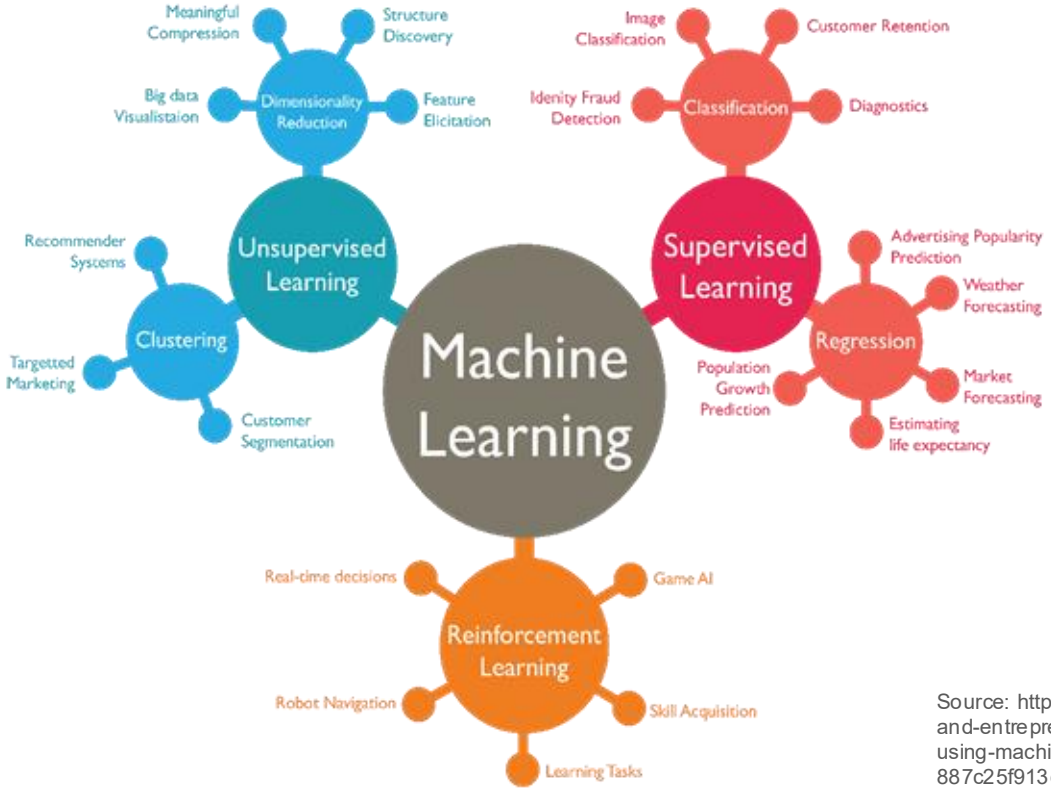
Source: https://www.linkedin.com/pulse/lets-understand-difference-between-machine-learning-vs-gauri-bapat

Recap: Machine Learning vs. Deep Learning vs AI



Source: https://www.ibm.com/blogs/systems/ai-machine-learning-and-deep-learning-whats-the-difference/

Recap: Types of Machine Learning



Source: https://medium.com/marketing-and-entrepreneurship/10-companies-using-machine-learning-in-cool-ways-887c25f913c3

Outline

- Statistics - Probability
- Typical Data Distribution
- Typical Measurements
  - Entropy, Cross Entropy
  - Mutual Information
  - Kullback-Leibler Divergence
- Learning Theory

What animal is it?



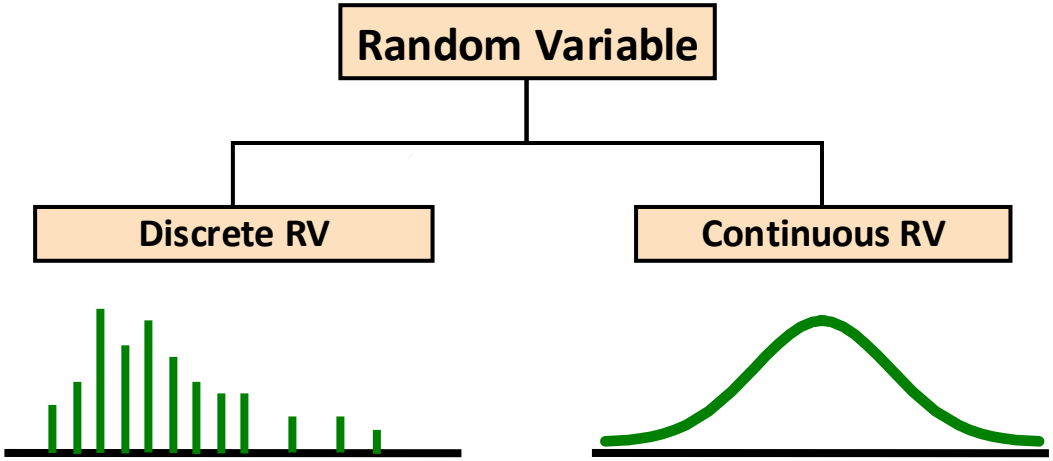
A cat?
or a tiger?

Game: Toss a coin



Random Variable (RV)

- Random variable (RV) is a variable whose possible values are numerical outcomes of a random phenomena.



Discrete Random Variable (DRV)

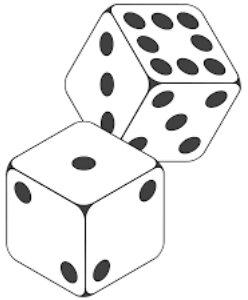
- Let toss the coin twice, and X is the number of the head occurs.

How many values of X ?



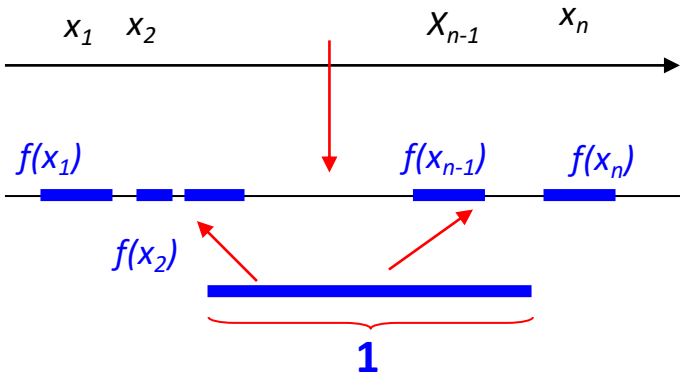
- Let toss the dice twice, and X is the sum of the dice

How many values of X?



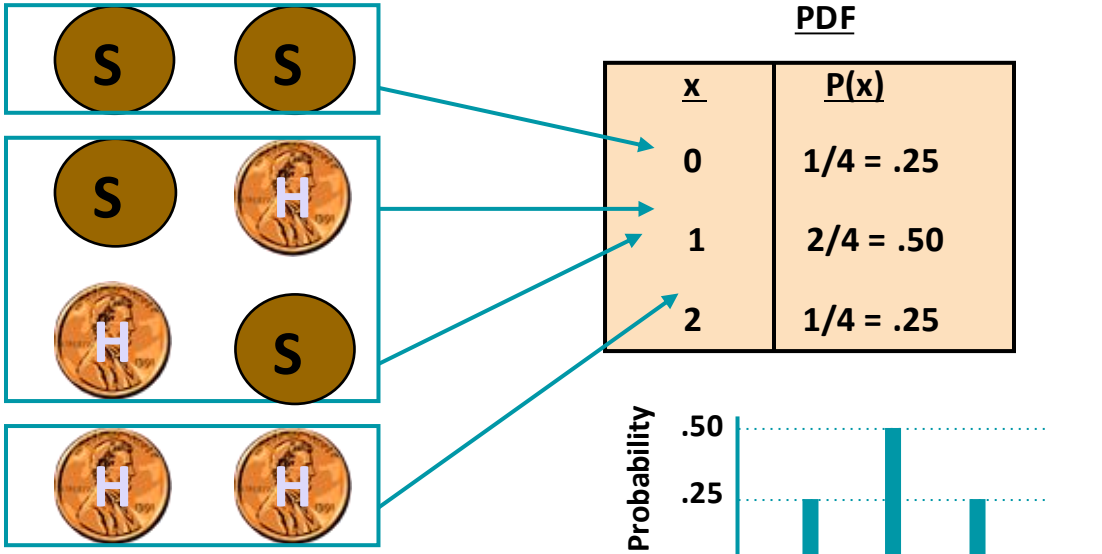
DRV - Probability Density Function (1)

- A discrete random variable X whose values in $x_1, x_2, \ldots, x_n$.
- Probability Density Function: $f(x_i) = P(X = x_i)$
- Denotation: $p_i = f(x_i) = P(X=x_i)$
- Conditions:
  - $f(x_i) \geq 0$
  - $\sum_{i=1}^{n} f(x_i) = 1$



DRV – Probability Density Function (2)

Example: Toss two coins, X is denoted as the occurrences of head.



There are 4 possible outcomes

Continuous Random Variable (CRV)

- Its value space is $\mathbb{R}$ or a subset of $\mathbb{R}$.



Weight & Height



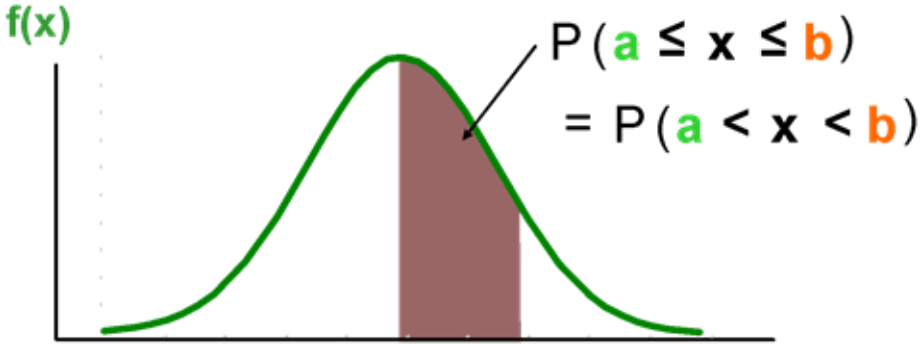
Time to
complete a task

CRV – Probability Density Function (1)

- f(x) is the probability density function of a given continuous random variable X, if:

$$i) f(x) \geq 0 \quad \forall x$$

$$ii) \int_{-\infty}^{+\infty} f(x)dx = 1$$



$$P(a < X < b) = \int_{a}^{b} f(x)dx$$

Cumulative Probability Distribution (1)

- Consider a random variable X, its cumulative probability function F(x) is defined as follows:

$$F(x) = P\left(X \leq x\right)$$

- The probability of X in (a,b] as:

$$P(a < X \leq b) = F(b) - F(a)$$

Cumulative Probability Distribution (2)

1) $0 \le F(x) \le 1$

2) F(x) is a non-decreasing function if ∀ a<b, we have F(a) ≤ F(b).

3) $F(-\infty) = \lim_{x \to -\infty} F(x) = 0$

$F(+\infty) = \lim_{x \to +\infty} F(x) = 1$

4) Probability density function f(x) = F'(x) as long as the derivative exits

Cumulative Probability Distribution (3)

- Discrete Random Variable:

$$F(x_0) = P(X \leq x_0) = \sum_{x_i \leq x_0} p_i$$

- Continuous Random Variable:

$$F(x_0) = P(X \leq x_0) = \int_{-\infty}^{x_0} f(x)dx$$

Expected Value of Random Variable

- Discrete Random Variable:

$$E[X] = \sum_{x} xP(X = x)$$

- Continuous Random Variable:

$$E[X] = \int_{-\infty}^{+\infty} xf(x)dx$$

Expected Value - Properties

1) E[C] = C, C: constant
2) E[CX] = C.E[X]
3) E[X + Y]=E[X] + E[Y]
4) E[XY] = E[X].E[Y] if X and Y are independent
5) Given a function h(x), we have

$$Eh(x) = \sum_{i=1}^{n} h(x_i) p_i$$ if X is discrete

$$Eh(x) = \int_{-\infty}^{+\infty} h(x) f(x) dx$$ if X is continuous

Variance of Random Variable

- Mean $\mu = \mathrm{E}[X]$
- Variance: $\mathrm{Var}[X] = \mathrm{E}\left[X - \mathrm{E}[X]\right]^2 = \mathrm{E}[X^2] - (\mathrm{E}[X])^2$
- Standard deviation: $\sigma = \sqrt{\sigma^2} = \sqrt{\mathrm{Var}[X]}$
- Discrete random variable:

$$\mathrm{Var}[X] = \sum_{i=1}^{n} (x_i - \mu)^2 p_i$$

- Continuous random variable:

$$\mathrm{Var}[X] = \int_{-\infty}^{+\infty} (x - \mu)^2 f(x) dx$$

Variance - Properties

1) $\mathrm{Var}[C] = 0$, C: constant
2) $\mathrm{Var}[CX] = C^2\,\mathrm{Var}[X]$

   $\mathrm{Var}[X + C] = \mathrm{Var}[X]$
3) $\mathrm{Var}[X + Y] = \mathrm{Var}[X] + \mathrm{Var}[Y]$ if X and Y are independent.

Three Rules of Probability

- Basic law

$$p(X = x) \geq 0, \quad \sum_{X=x} p(X = x) = 1.$$

- Sum rule

$$p(X) = \sum_Y p(X, Y).$$

- Product rule

(Bayesian Rule)

$$p(X, Y) = p(Y|X)p(X).$$

Joint Probability

$$\text{both are discrete} \qquad \sum_{x,y} p(x,y) = 1$$

$$\text{both are continuous} \qquad \int p(x,y)dxdy = 1$$

$$x \text{ is discrete, } y \text{ is continuous} \qquad \sum_{x} \int p(x,y)dy = \int \left(\sum_{x} p(x,y) \right) dy = 1$$

Marginal Probability

- Discrete random variable:

$$p(x) = \sum_y p(x, y)$$

$$p(y) = \sum_x p(x, y)$$

- Continuous random variable:

$$p(x) = \int p(x, y) dy$$

$$p(y) = \int p(x, y) dx$$

Conditional Probability

- Bayesian Rule: $p(y|x)p(x) = p(x|y)p(y)$

- Conditional Probability:

$$p(x|y=9) == \frac{p(x,y=9)}{\sum_x p(x,y=9)} = \frac{p(x,y=9)}{p(y=9)}$$

Posterior-, Prior Probability & Likelihood

- A random variable X could be predicted via the parameters θ:

$$p(\theta|X) = \frac{p(X|\theta)}{p(X)} p(\theta)$$



$$p(\theta|X) \propto p(X|\theta) \times p(\theta)$$

Posterior probability

Likelihood

Prior probability

Independence & Dependence

- If x, y are independent, so:

$$p(x|y) = p(x)$$
$$p(y|x) = p(y)$$

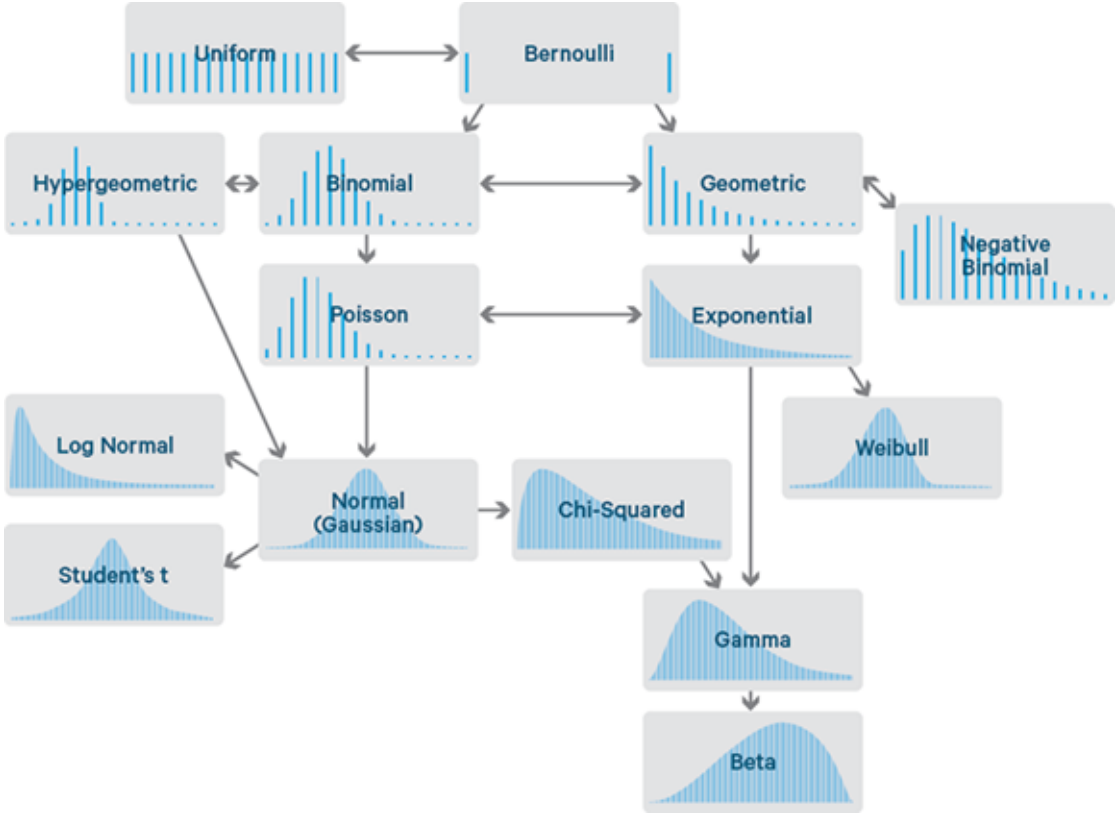
- Bayesian Rule:

$$p(x, y) = p(x|y)p(y) = p(x)p(y)$$

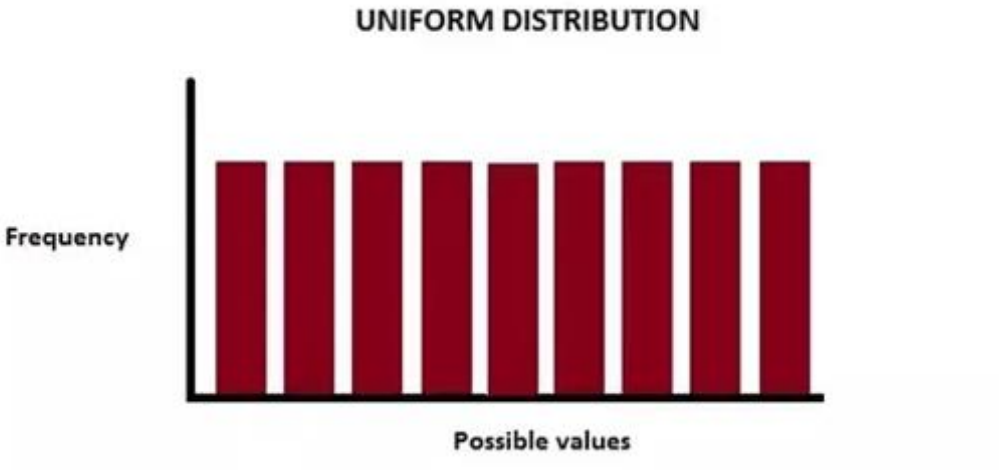
Outline

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Typical Data Distributions



Uniform Distribution



P(X) = 1 / total number of possible outcomes

Uniform Distribution – Example

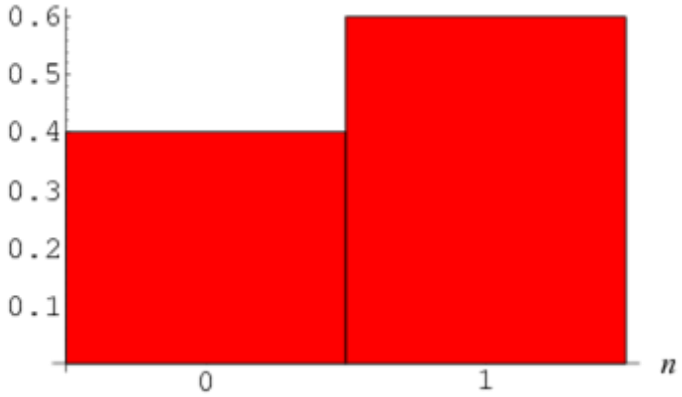


The probability of doomsday being Monday is 1/7



The probability that a numeric letter appearing in 500k VNĐ note is 1/10

Bernoulli Distribution



$$=> \text{Bern}_x[0.6]$$

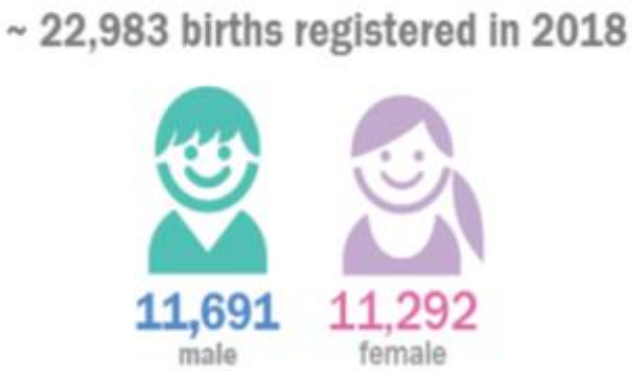
$$P(x) = \text{Bern}_x[\lambda] = \lambda^x(1-\lambda)^{1-x}$$

$$x \in \{0,1\}, \lambda \in [0,1]$$

Bernoulli Distribution – Example

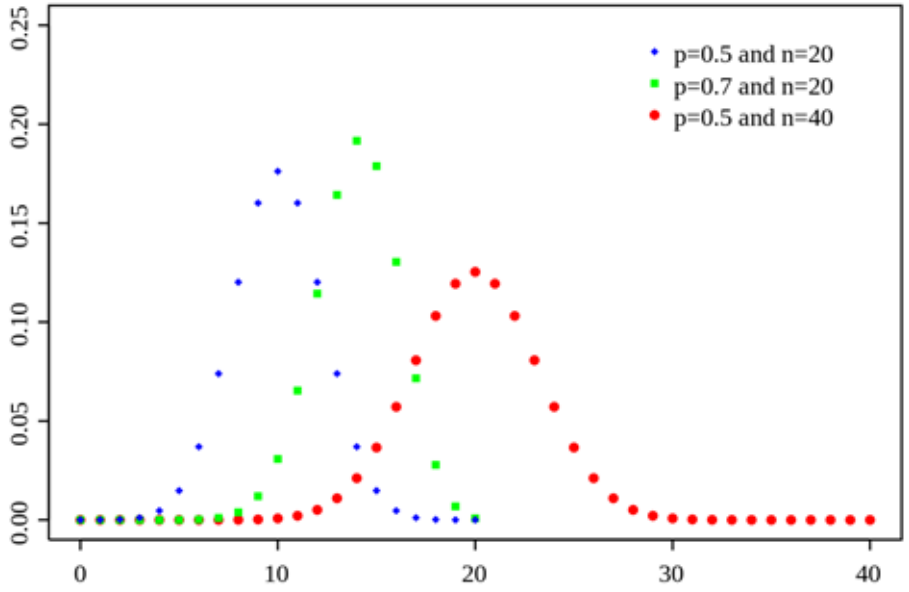


The probability of head/tail $\lambda \approx 0.5$



The probability of male/female baby

Binary Distribution



$$\mathbf{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k} \quad \text{voi } k = 0, 1, 2, \ldots, n$$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

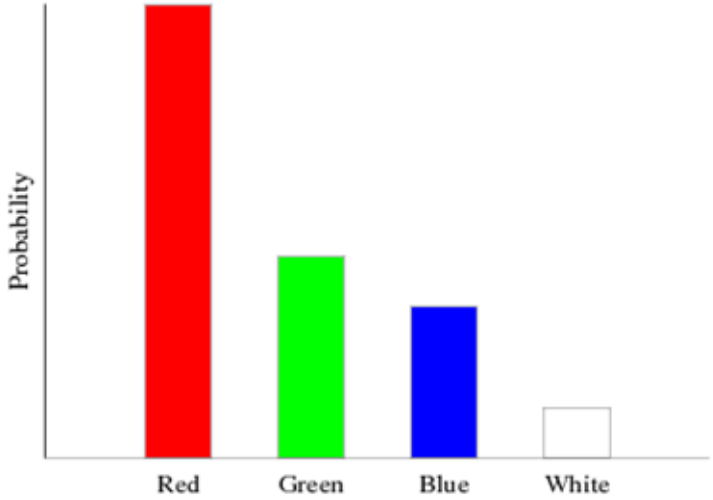
Binary Distribution – Example



The probability getting X times of head in 10 times tossing a coin

X	P(X)
0	0.000977
1	0.009766
2	0.043945
3	0.117188
4	0.205078
5	0.246094
6	0.205078
7	0.117188
8	0.043945
9	0.009766
10	0.000977

Categorical Distribution

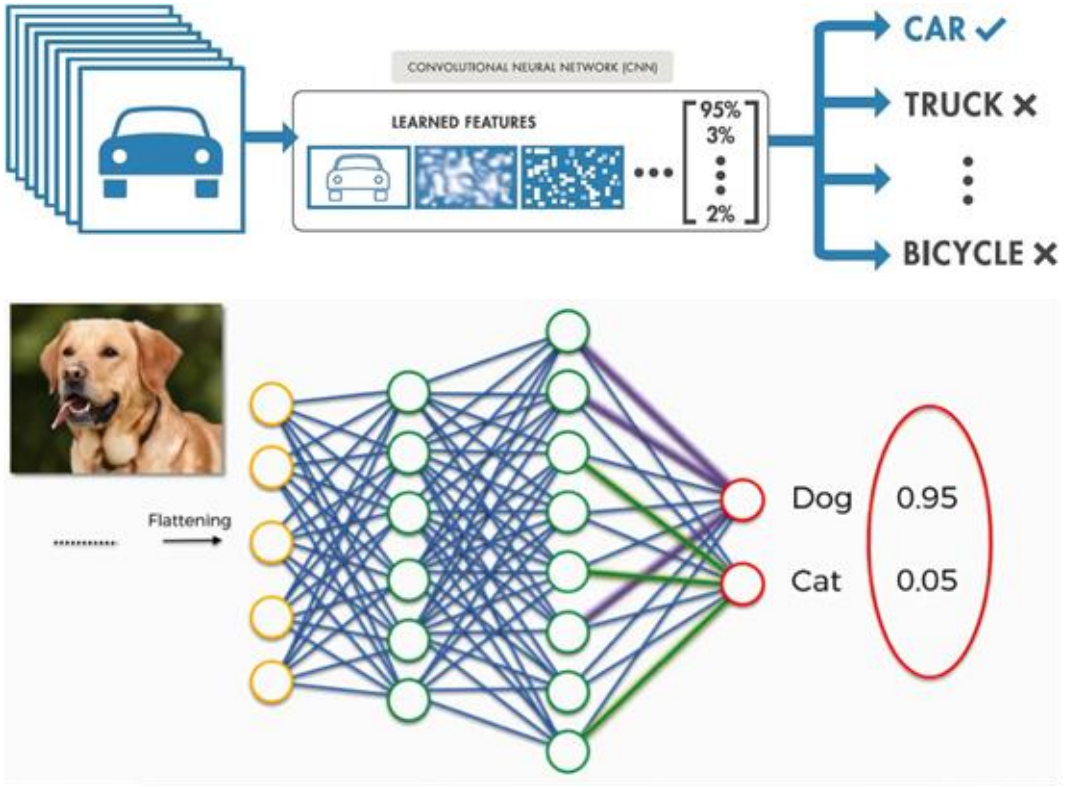


$$P(x) = \mathrm{Cat}_x[\lambda]$$

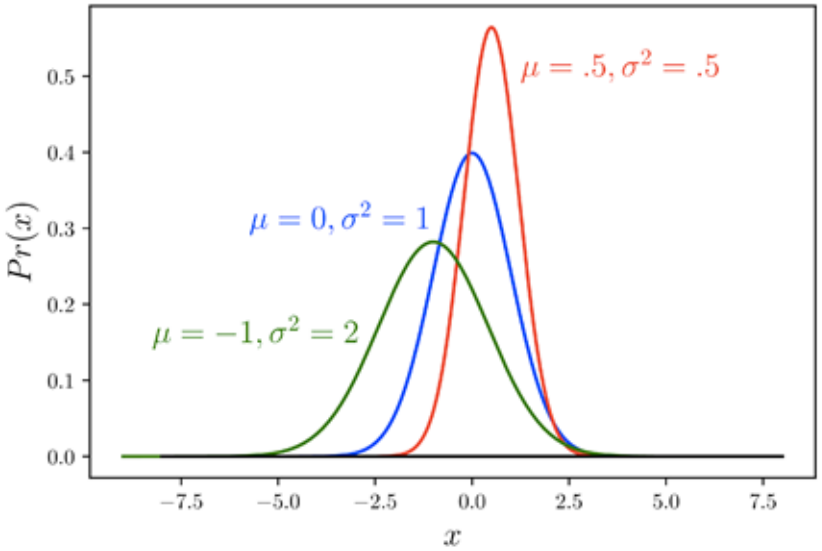
$$\lambda = \{\lambda_1, \ldots, \lambda_{|C|}\}; \lambda_k \in [0,1]$$

It is general form of Bernoulli distribution with more than 2 possible outcomes

Categorical Distribution - Example



Univariate Normal Distribution

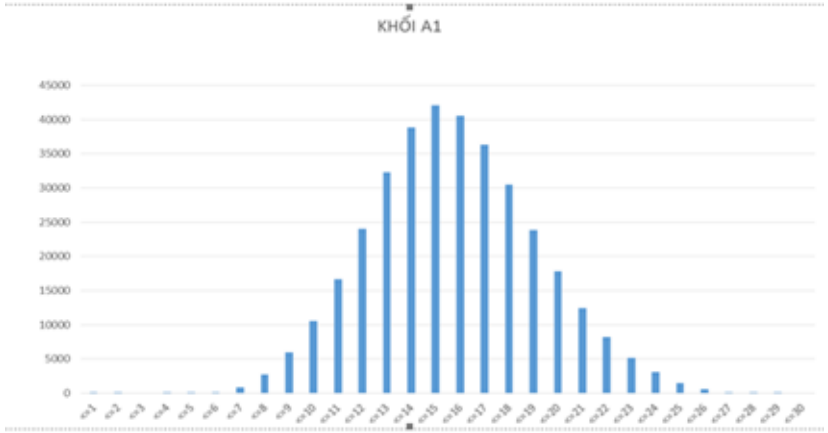


$$P(X) = \text{Norm}_x[\mu, \sigma^2] = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

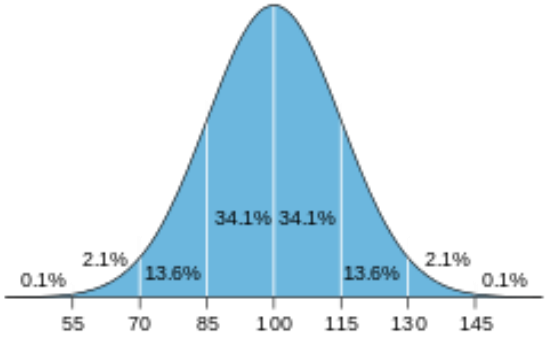
Mean μ, Variance σ^2

Normal distribution = Gaussian distribution

Univariate Normal Distribution - Example

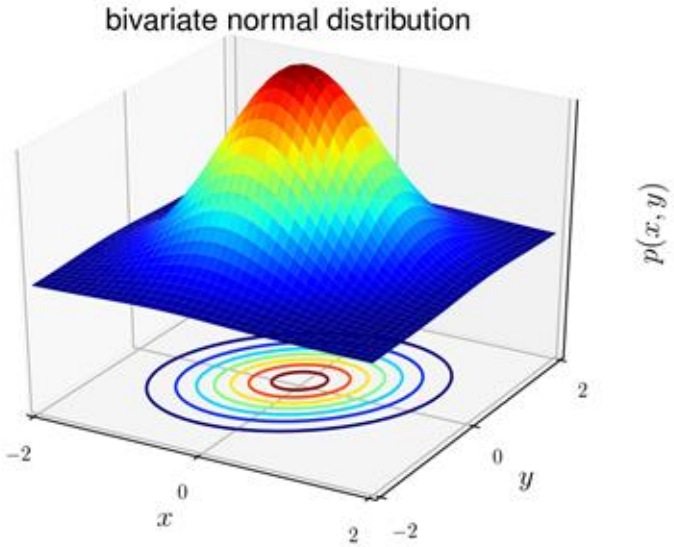


Score Distribution of a University Entrance Exam



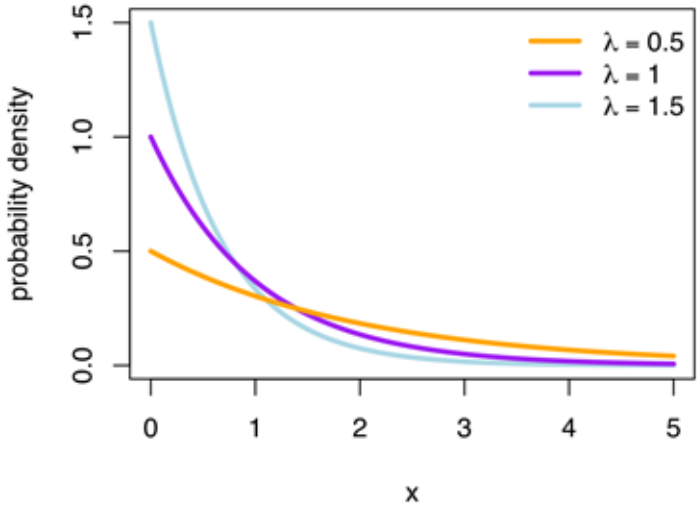
IQ Distribution

Multivariate Normal Distribution



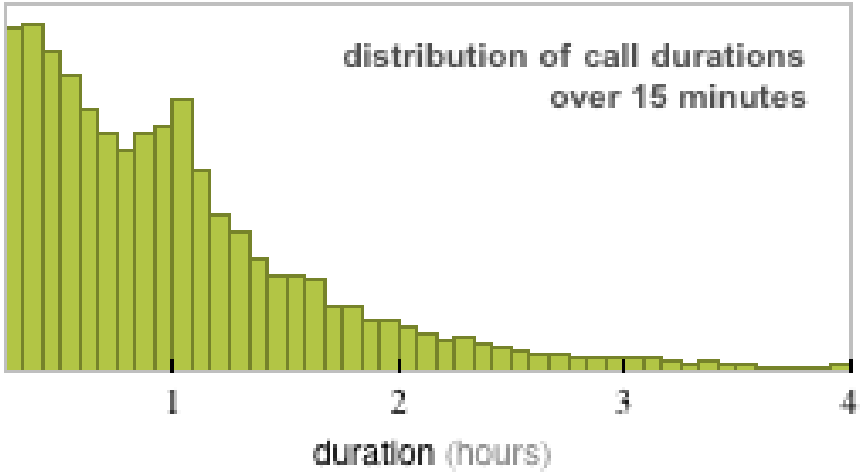
$$p(\mathbf{x}) = \frac{1}{(2\pi)^{D/2}|\Sigma|^{1/2}} \exp\left(\frac{1}{2}(\mathbf{x}-\mu)^T\Sigma^{-1}(\mathbf{x}-\mu)\right)$$

Exponential Distribution



$$P(x) = \text{Exp}(\lambda) = \lambda e^{-\lambda x} \quad x \geq 0$$

Exponential Distribution - Example



Time distribution of call duration over 15 minutes in a Telecom company

Outline

- Statistics - Probability
- Typical Data Distribution
- Typical Measurements
  - Entropy, Cross Entropy
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- Learning Theory

Typical Measurement - Entropy

- Measure the ‘uncertainty’ or ‘surprise’ in data
- Discrete random variable:

$$\mathrm{H}(X) = -\sum_{i=1}^{n} \mathrm{P}(x_i) \log \mathrm{P}(x_i)$$

- Continuous random variable:

$$h(X) = -\int_{\mathcal{X}} f(x) \log f(x)\, dx$$

Typical Measurement – Mutual Information

- Measure the mutual dependence between two random variables.

$$
\begin{aligned}
\mathrm{I}(X;Y) &\equiv \mathrm{H}(X) - \mathrm{H}(X|Y) \\
&\equiv \mathrm{H}(Y) - \mathrm{H}(Y|X) \\
&\equiv \mathrm{H}(X) + \mathrm{H}(Y) - \mathrm{H}(X,Y) \\
&\equiv \mathrm{H}(X,Y) - \mathrm{H}(X|Y) - \mathrm{H}(Y|X)
\end{aligned}
$$

Typical Measurement – Cross Entropy

- Measure the difference between two distributions
- Discrete random variable:

$$H(p, q) = -\sum_{x \in \mathcal{X}} p(x) \log q(x)$$

- Continuous random variable:

$$H(p, q) = -\int_{\mathcal{X}} P(x) \log Q(x) \, dr(x)$$

Typical Measurement – KL Divergence

- Measure the difference between two distributions
- Discrete random variable:

$$D_{\mathrm{KL}}(P \parallel Q) = -\sum_{x \in \mathcal{X}} P(x) \log\left(\frac{Q(x)}{P(x)}\right)$$

- Continuous random variable:

$$D_{\mathrm{KL}}(P \parallel Q) = \int_{-\infty}^{\infty} p(x) \log\left(\frac{p(x)}{q(x)}\right) dx$$

Quiz

1. What is the probability that the total of two dice will be greater than 9, given that the first dice is a 5?

2. Suppose there is a school having 60% boys and 40% girls as students. The girls wear trousers or skirts in equal numbers; all boys wear trousers. An observer sees a (random) student from a distance; all the observer can see is that this student is wearing trousers. What is the probability this student is a girl?

Quiz

3. Suppose we want to predict the next day is Rain or Sunny. We can modeling the target distribution by which distribution?

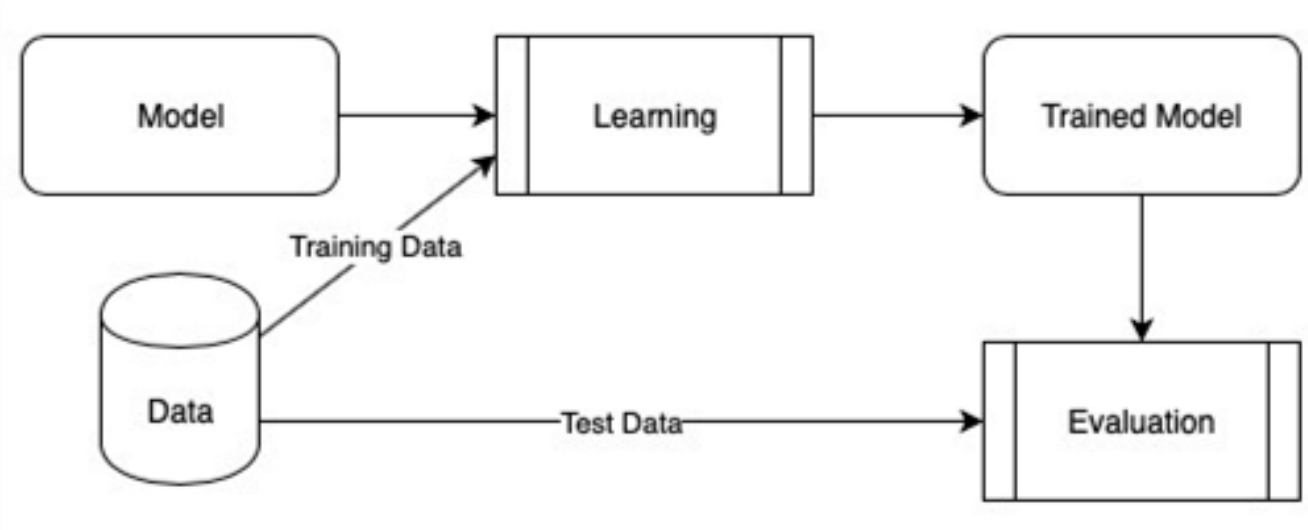
4. Suppose we want to predict the GPA of a UET-student, which distribution can be used to model the GPA

5. What happen to the KL divergence $D_{KL}(P||Q)$ if the support of P and Q are not overlapped

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Inductive Learning



The Statistical Learning Framework

The formal definition of the learner's input:

- **Domain Set**: An arbitary set $\mathcal{X}$. $\mathcal{X}$ represents the space contain the objects we want to learn
- **Label Set**: A set of definitions $\mathcal{Y}$ which defines on $\mathcal{X}$. In the binary case, think of $\{0, 1\}$ or $\{-1, 1\}$ or $\{class_1, class_2, class_3\}$ or a *Real Value* if the task is regression task. It can be extended to n classes.
- **Training data**: A sequence $S = \{(x_1, y_1), \ldots, (x_m, y_m)\}$ in the domain $\mathcal{X} \times \mathcal{Y}$.

In general, S is the specific definition of the experience E in **supervised classification**.

The Statistical Learning Framework

The formal definition of the learner's output:

- A prediction rule:

$$h : \mathcal{X} \to \mathcal{Y} \tag{1}$$

- h: classifier, predictor, hypothesis or mapping function. For example, h is the linear function with thresholding in previous lesson.
- Given an algorithm A, we denote $A(S)$ is the set of classifier/hypothesis generated by apply A on S

h could be more general than the definition of a function
One example is: h is the stochastic function

The Statistical Learning Framework

A simple **data-generation pipeline**:

- Assume we have a probability distribution $\mathcal{D}$ over $\mathcal{X}$,
- Given $\mathcal{D}$, we draw a sample $x_i \sim \mathcal{D}$
- Assume we have a **correct** label function c:

$$c : \mathcal{X} \to \mathcal{Y} \tag{2}$$

that $y_i = c(x_i)$, and $S = ((x_1, y_1), \ldots, (x_m, y_m))$

Notes:

1. c is function we want to recover. However, we do not know $\mathcal{D}$ and c, our learning algorithm A can only access $\mathcal{X}$, $\mathcal{Y}$, S
2. Taken into account that, every time the **data-generation pipeline** runs, we can get a new training data S

The Statistical Learning Framework

Given domain set $\mathcal{X}$ and label set $\mathcal{Y}$, the data distribution $\mathcal{D}$ and the labeling function c.

The error of a prediction rule $h : \mathcal{X} \to \mathcal{Y}$ is defined as:

$$L_{\mathcal{D},c}(h) \stackrel{\text{def}}{=} \mathbb{P}_{x \sim \mathcal{D}}[h(x) \neq c(x)] \tag{3}$$

However, we do not know $\mathcal{D}$ and c in practice

The Statistical Learning Framework

The Empirical Risk Minimization (ERM) framework learns a learner

$$h_S : \mathcal{X} \to \mathcal{Y}$$

that minimize the **the training error** without knowing $\mathcal{D}$ and c.

The **training error** is defined as

$$L_S(h) \overset{\text{def}}{=} \frac{|\{i \in [m] : h(x_i) \neq y_i\}|}{m} \tag{4}$$

Note that: we use h_S to emphasize our learners depends on S, a specific S

The Statistical Learning Framework

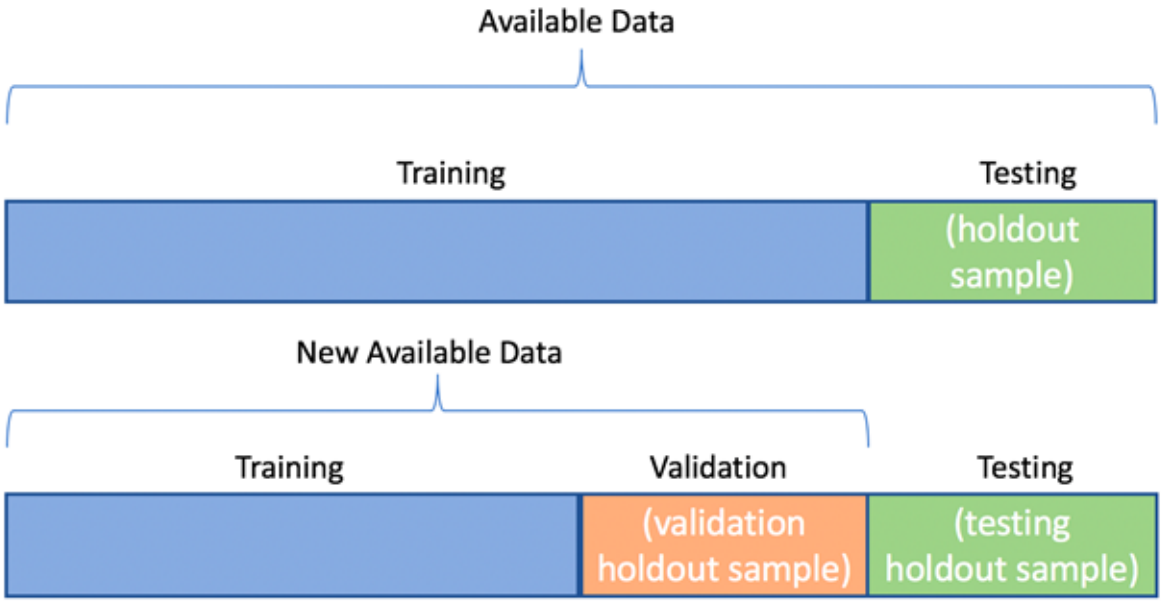
Assume we have two dataset S_1, S_2, which both sample from $\mathcal{D}$
We have the error on S_1, S_2 is defined as:

$$L_{S_1}(h) \stackrel{\text{def}}{=} \frac{|\{(x_i, y_i) \in S_1, i \in [m_1] : h(x_i) \neq y_i\}|}{m_1} \tag{5}$$

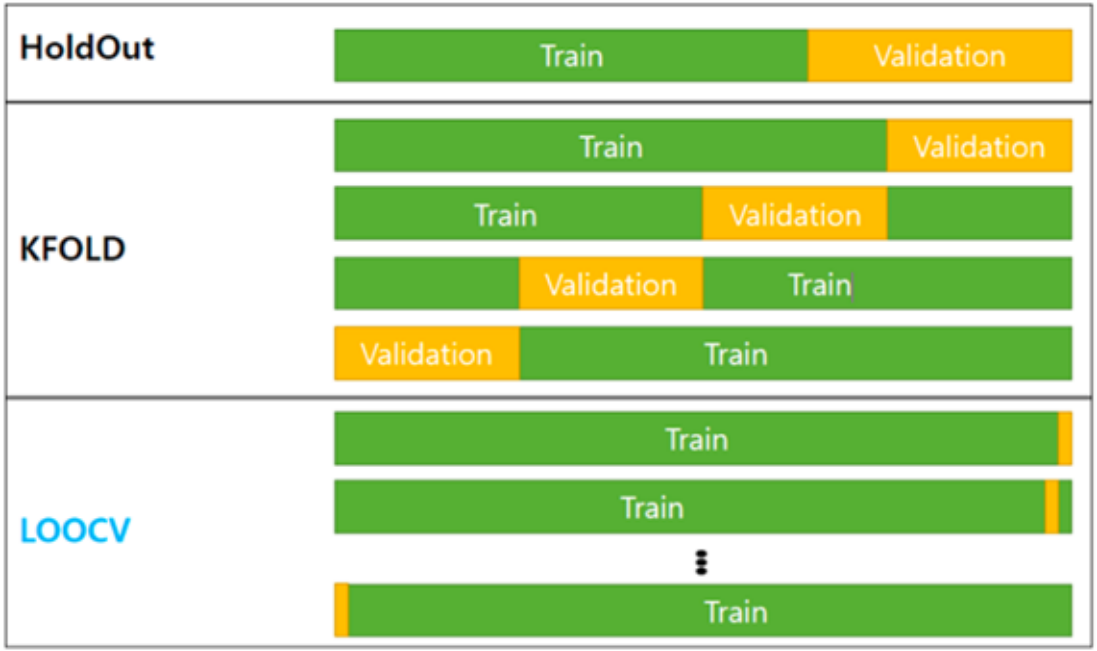
$$L_{S_2}(h) \stackrel{\text{def}}{=} \frac{|\{(x_i, y_i) \in S_2, i \in [m_2] : h(x_i) \neq y_i\}|}{m_2} \tag{6}$$

1. Overfitting: we have $L_{S_1}(h)$ near 1 and $L_{S_2}(h)$ near 0.5
2. Generalization: we have $L_{S_1}(h) \sim L_{S_1}(h)$

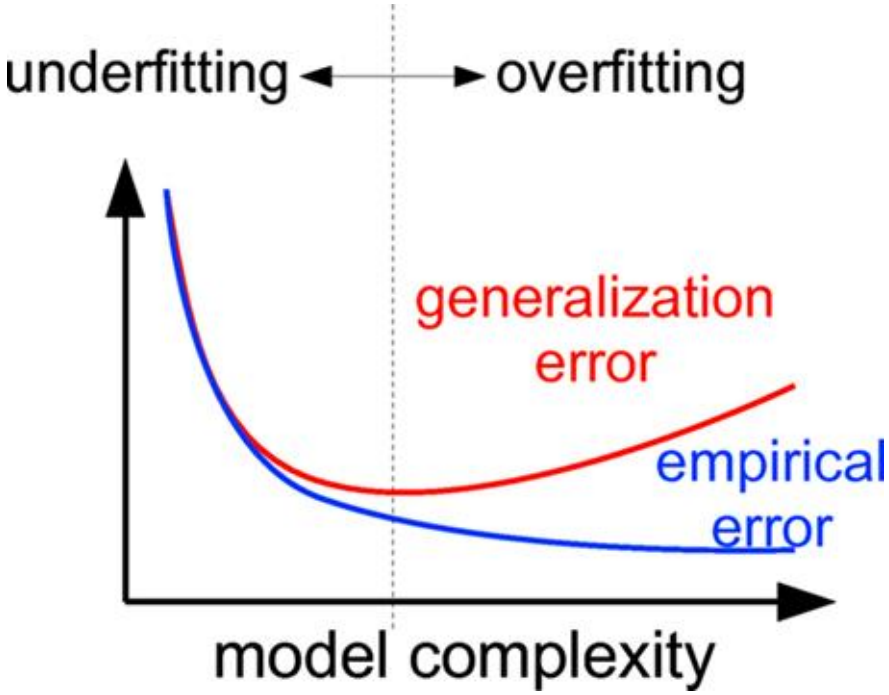
Train/Test/Validation



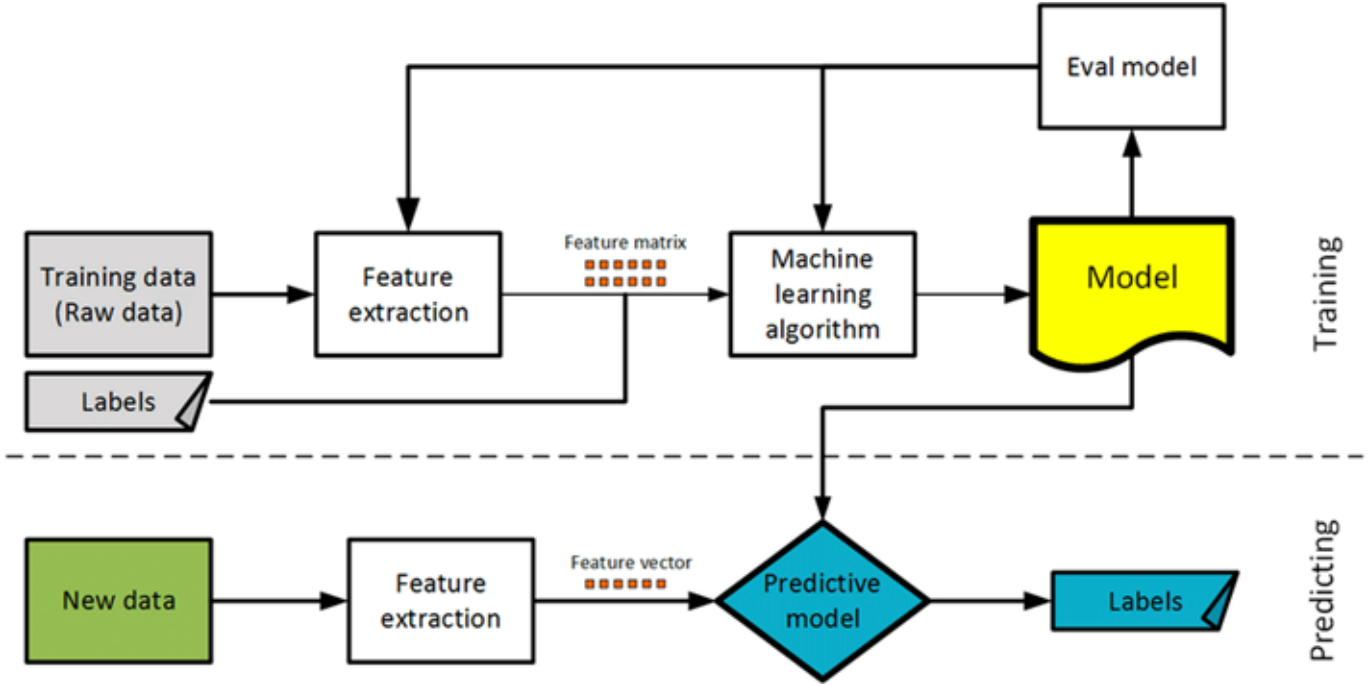
Cross Validation



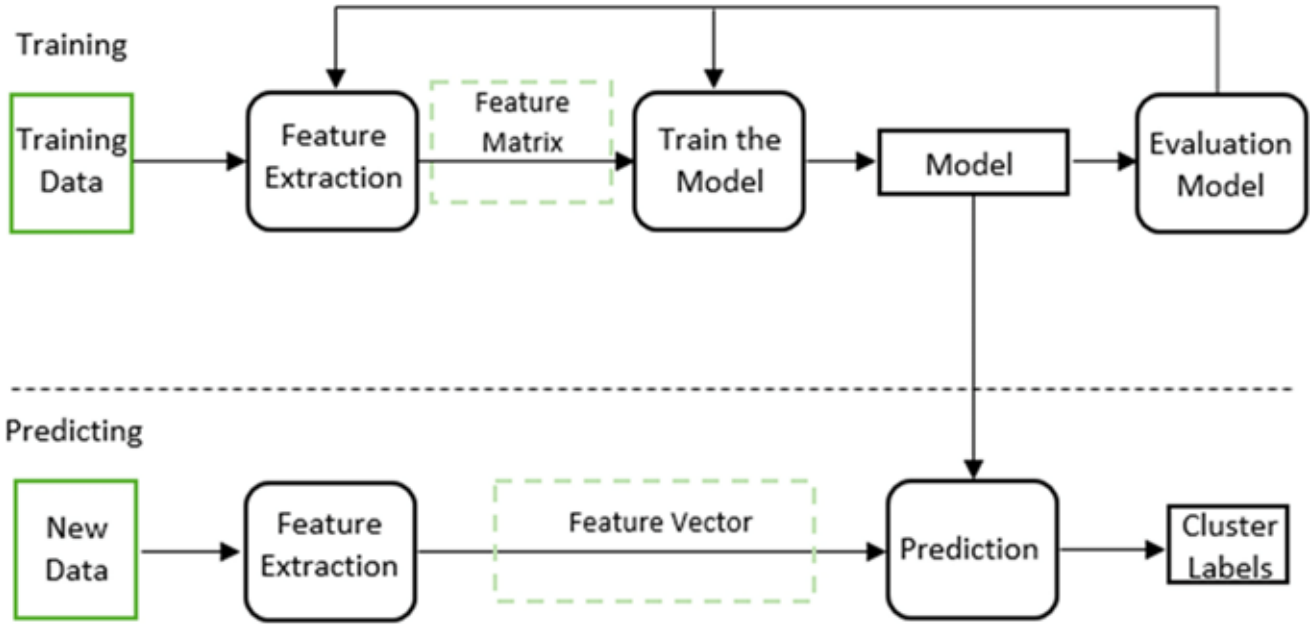
Generalization- & Empirical Error



Supervised Learning Workflow



Unsupervised Learning Workflow



Summary

- Statistics - Probability
- Typical Data Distribution
- Typical Measurements
  - Entropy, Cross Entropy
  - Mutual Information
  - Kullback-Leibler Divergence
- Learning Theory

Summary

Reading:

Chapter 2 and Chapter 3 of PML.

For next lectures: 3.1 to 3.2, ESL, 11.2 PML

Homework: Parameter Estimation

$X \in \{0, 1\}$ with probability $P(X = 1) = \theta$, written as $X \sim Ber(\theta)$.
We also have $P(X = 0) = 1 - \theta$.

- A biased coin: $\theta =$ probability of head
- Binary classification: $P(y|x) = Ber(\theta(x))$
  $\rightarrow$ probability of class 1 is a function of input

Homework: Parameter Estimation

Toss a coin (sampling) N times, the number of times heads come up (number 1) is s times, what is the parameter θ of the coin (Bernoulli distribution)?
An intuitive guess: $\theta = \frac{s}{N}$, why does this number make sense?
Let $x_i \in \{0, 1\}$ is the values from the ith toss.
The probability of the data $D = \{x_1, x_2, \ldots, x_N\}$ under the model $X \sim Ber(\theta)$ is

$$L(\theta) = P(D) = P(x_1, x_2, \ldots, x_N) = \prod_{i=1}^{N} P(x_i)$$
$$= \prod_{i=1}^{N} \theta^{x_i}(1-\theta)^{1-x_i}$$

Homework: Parameter Estimation

$L(\theta)$ is the *likelihood* of θ with respect to the dataset D
MLE: Find θ for which $L(\theta)$ is maximized.

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^{n} x_i \log \theta + (1 - x_i) \log(1 - \theta)$$

$$\ell'(\theta) = \sum_{i=1}^{n} x_i/\theta - (1 - x_i)/(1 - \theta) = 0$$

$$\frac{1}{\theta} \underbrace{\sum_{i=1}^{n} x_i}_{s} = \frac{1}{1 - \theta} \underbrace{\sum_{i=1}^{n} (1 - x_i)}_{N-s}$$

$$s(1 - \theta) = (N - s)\theta$$

$$\theta^{MLE} = \frac{s}{N}$$

Homework: Parameter Estimation

- Unbiased: $\mathbb{E}[\theta^{MLE}] = \theta$
- Variance goes to 0: $\mathbb{V}[\theta^{MLE}] = \theta(1-\theta)/N$
- Consistent: $\mathbb{P}\{|\theta^{MLE} - \theta| \geq \epsilon\} \overset{n \to \infty}{\longrightarrow} \infty$

Thank you